

## M002 Week 6 — Redstone & Levers

**Topic:** Redstone Levers · **Course:** Maze Madness · **Level:** Advanced · **Time:** about 55 minutes

This week the maze has **redstone levers**. The agent detects redstone, flips a lever with `interact` to open a closed redstone door, then walks through to clear the maze.

### 1 · The API

| Code                                         | What it does              |
|----------------------------------------------|---------------------------|
| <code>agent.detect(REDSTONE, FORWARD)</code> | True if redstone is ahead |
| <code>agent.interact(...)</code>             | flip the lever in front   |
| <code>agent.move(FORWARD, 1)</code>          | step one block            |
| <code>agent.turn(RIGHT_TURN)</code>          | turn right                |
| <code>player.on_chat("go", run)</code>       | run code on chat word     |

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## 2 · Predict 🧠

```
agent.move(FORWARD, 1)
agent.interact(...)
agent.move(FORWARD, 1)
```

**The agent reaches a lever and flips it. What happens to the door?**

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```
agent.interact(...)
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
```

**There is no lever in front. What happens?**

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### 3 · Trace the Code 🔍

```
while not agent.detect(REDSTONE, FORWARD):  
    agent.move(FORWARD, 1)  
agent.interact(...)  
agent.turn(RIGHT_TURN)  
agent.move(FORWARD, 1)  
agent.move(FORWARD, 1)
```

**Describe what the agent does, step by step, in plain English.**

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## 4 · Spot and Fix the Bugs 🐛

Each block is broken. Write the fixed code, then say why.

**Bug A** — Flip the lever **first**, then walk through the open door. The lever is at the first wall.

```
agent.move(FORWARD, 1)
agent.move(FORWARD, 1)
agent.interact(...)
```

**Fixed code:**

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**Why was it wrong?**

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**Bug B** — Walk **up to** the lever, then flip it **once**. This flips on every step.

```
while not agent.detect(REDSTONE, FORWARD):
    agent.move(FORWARD, 1)
    agent.interact(...)
```

**Fixed code:**

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**Why was it wrong?**

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**Bug C** — After flipping, the agent must **turn around** and go back through the door. The door is behind it.

```
agent.interact(...)
agent.move(FORWARD, 1)
```

**Fixed code:**

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**Why was it wrong?**

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## 5 · Write the Code 🖋️

The agent starts in the maze. Somewhere ahead is a lever. Write code that walks to the lever, flips it, and walks through the open door.

```
def run():
    # your code here

player.on_chat("go", run)
```

## 6 · Finish the Maze 📷

Open your homework world. The maze has a **lever** that opens a **closed door**. Write code that:

1. moves the agent to the lever,
2. flips it with `interact`,
3. walks through the open door,
4. reaches the goal.

Send a photo or video of the agent at the end. Then write 4 to 6 sentences.

First, I walked the agent to ...

I used `interact` to ...

Once the door opened, the agent ...

The hardest part was ...

To fix it, I ...

If I had more time, I would ...

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## 7 · Record Your Walkthrough 🎥

Film the agent running the maze on a phone. Teach it like the viewer has never seen it. Use these words: **lever, interact, flip, door, open**.

1. Show the closed door and the lever.
2. Run your code. Show the agent flip the lever.
3. Read your code out loud. Say which line opens the door.
4. Show one bug you hit and how you fixed it.

**Plan what you will say before you record. You can read it while filming.**

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**Submit** ✓

Send this worksheet + your walkthrough video to teacher on KakaoTalk.